Compact Spherical FLRW Residuals: Closure, Local ΛCDM Degeneracy, and Amplitude-Selection Boundaries

Andrei Preece[1, *] and Boris Batenin[1, †]

[1] *Independent Researcher, Berlin, Germany*

(Dated: May 15, 2026)

For a fixed spacetime metric and a declared visible-sector subtraction, the Einstein residual

$$T^{\mathcal{D}}_{\mu\nu} = \frac{c^4}{8\pi G} G_{\mu\nu} - T^{\rm vis}_{\mu\nu}$$

is covariantly fixed. We prove that, within the homogeneous locally-FLRW residual class, this residual has the minimal pressureless-current plus fixed-four-form stress-tensor form, at the homogeneous stress-tensor level, precisely when

$$\frac{\mathrm{d}}{\mathrm{d}\ln a}[-p_{\mathcal{D}}] = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}\ln a}\left\{a^3(\epsilon_{\mathcal{D}} + p_{\mathcal{D}})\right\} = 0,$$

with the corresponding constants nonnegative. Equivalently,

$$\epsilon_{\mathcal{D}}(a) = \epsilon_{\mathrm{w},0} a^{-3} + \epsilon_q, \qquad p_{\mathcal{D}}(a) = -\epsilon_q.$$

Thus the minimal homogeneous branch is locally identical to a cold-dark-matter plus cosmological-constant residual with the same amplitudes and curvature. Compact spherical topology may still affect global volume, quotient-projected spectra, compact covariance, and observer-position dependence for non-globally-homogeneous quotients, but it does not by itself determine Ω_c, Ω_Λ, or $\Lambda_{\rm eff}$. The result is therefore a residual-closure theorem and an amplitude-selection boundary, not an abundance-origin theorem or a topology detection.

I. CLAIM BOUNDARY

This paper proves a conditional stress-tensor closure theorem. It does not derive the observed dark-sector abundances, does not select a flux branch, and does not claim detection of compact topology. Once the metric $g_{\mu\nu}$, the Einstein-side convention, and the visible-sector subtraction $T^{\rm vis}_{\mu\nu}$ are fixed, the Einstein residual is fixed. If this residual is homogeneous and locally FLRW, then it has the minimal pressureless-current plus fixed-vacuum stress-tensor form if and only if

$$-p_{\mathcal{D}} = \text{constant} \geq 0, \qquad a^3(\epsilon_{\mathcal{D}} + p_{\mathcal{D}}) = \text{constant} \geq 0. \tag{1}$$

Microscopic current and four-form realizations may impose further amplitude selection rules; the closure theorem itself does not supply them. The corresponding stress tensor is locally indistinguishable from cold dark matter plus Λ with the same amplitudes and curvature. Any distinction must therefore come from compact spectral projection, quotient covariance, perturbative null tests, or an additional amplitude-selection rule.

The paper is complete only at this declared level: homogeneous residual closure, compact-background bookkeeping, perturbative null conditions, and amplitude-selection boundaries. It is not a compact Boltzmann solver, a likelihood analysis, or a microscopic dark-sector completion.

* andre.preece@web.de

† batenin@web.de

II. EINSTEIN RESIDUAL AND CONVENTION COVARIANCE

Let $(M, g_{\mu\nu})$ be a four-dimensional spacetime, and let $T^{\rm vis}_{\mu\nu}$ denote the stress tensor assigned to the declared visible sector. Define

$$\boxed{T^{\mathcal{D}}_{\mu\nu} := \frac{c^4}{8\pi G} G_{\mu\nu} - T^{\rm vis}_{\mu\nu}.} \tag{2}$$

A bare cosmological-constant term is included in $T^{\mathcal{D}}_{\mu\nu}$ unless the Einstein-side convention is explicitly changed to put $G_{\mu\nu} + \Lambda g_{\mu\nu}$ on the left-hand side.

Proposition 1 (Residual uniqueness and convention covariance)**.** *If*

$$G_{\mu\nu} = \frac{8\pi G}{c^4}\left(T^{\rm vis}_{\mu\nu} + S_{\mu\nu}\right), \tag{3}$$

then $S_{\mu\nu} = T^{\mathcal{D}}_{\mu\nu}$. If the visible subtraction is changed by $X_{\mu\nu}$, then

$$T^{\rm vis}_{\mu\nu} \to T^{\rm vis}_{\mu\nu} + X_{\mu\nu}, \qquad T^{\mathcal{D}}_{\mu\nu} \to T^{\mathcal{D}}_{\mu\nu} - X_{\mu\nu}. \tag{4}$$

If the Einstein side is changed from $G_{\mu\nu}$ to $G_{\mu\nu} + \Lambda_0 g_{\mu\nu}$ while the visible subtraction is kept fixed, then

$$T^{\mathcal{D}}_{\mu\nu} \to T^{\mathcal{D}}_{\mu\nu} + \frac{c^4 \Lambda_0}{8\pi G} g_{\mu\nu}. \tag{5}$$

We use standard differential-geometric conventions [1]. The contracted Bianchi identity gives

$$\nabla^\mu T^{\mathcal{D}}_{\mu\nu} = -\nabla^\mu T^{\rm vis}_{\mu\nu}. \tag{6}$$

Thus separate residual conservation is not automatic; it follows only when the visible sector is separately conserved in the adopted convention. This point matters because the minimal dust-plus-vacuum source branch is separately conserved.

We use a timelike four-velocity U^μ normalized by $U^\mu U_\mu = -c^2$ and the projector

$$h_{\mu\nu} = g_{\mu\nu} + \frac{U_\mu U_\nu}{c^2}. \tag{7}$$

For a perfect fluid with physical energy density ϵ and pressure p,

$$T_{\mu\nu} = \epsilon \frac{U_\mu U_\nu}{c^2} + p h_{\mu\nu}. \tag{8}$$

III. LOCALLY-FLRW RECONSTRUCTION

On the round cover, use

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + a^2(t) A_0^2 \mathrm{d}\Omega_3^2, \qquad \Omega_K = -\frac{c^2}{H_0^2 A_0^2} < 0, \tag{9}$$

where A_0 is the present curvature radius of the simply connected cover. For a quotient $S^3_{A_0}/\Gamma$, the local metric is unchanged while the volume and global spectrum are restricted. Let

$$E(a) = \frac{H(a)}{H_0}, \qquad \epsilon_{\mathrm{crit},0} = \frac{3H_0^2 c^2}{8\pi G}. \tag{10}$$

The local Friedmann equations reconstruct a homogeneous residual as

$$\epsilon_{\mathcal{D}}(a) = \epsilon_{\mathrm{crit},0}\left[E^2(a) - \Omega_K a^{-2}\right] - \epsilon_{\mathrm{vis}}(a), \tag{11}$$

$$p_{\mathcal{D}}(a) = -\frac{\epsilon_{\mathrm{crit},0}}{3}\left[2E(a)\frac{\mathrm{d}E}{\mathrm{d}\ln a} + 3E^2(a) - \Omega_K a^{-2}\right] - p_{\mathrm{vis}}(a). \tag{12}$$

These are reconstruction identities, not equations of microscopic origin.

Theorem 2 (Locally-FLRW residual reduction)**.** *After a fixed visible subtraction, every homogeneous and isotropic symmetric residual tensor is uniquely of the form*

$$T^{\mathcal{D}}_{\mu\nu} = \epsilon_{\mathcal{D}}(a)\frac{U_\mu U_\nu}{c^2} + p_{\mathcal{D}}(a) h_{\mu\nu}. \tag{13}$$

If the residual is separately conserved, then

$$\frac{\mathrm{d}\epsilon_{\mathcal{D}}}{\mathrm{d}\ln a} + 3(\epsilon_{\mathcal{D}} + p_{\mathcal{D}}) = 0. \tag{14}$$

Proof. Local homogeneity and isotropy leave only $U_\mu U_\nu$ and the spatial projector $h_{\mu\nu}$ as invariant symmetric rank-two tensors. Taking the covariant divergence in FLRW gives Eq. (14). □

IV. CLOSURE-PAIR ATLAS AND MINIMAL BRANCH

Define the closure variables

$$\boxed{\mathcal{A}_q(a) := -p_{\mathcal{D}}(a), \qquad \mathcal{A}_{\mathrm{w}}(a) := a^3[\epsilon_{\mathcal{D}}(a) + p_{\mathcal{D}}(a)].} \tag{15}$$

They are a lossless reparametrization of the homogeneous residual:

$$p_{\mathcal{D}}(a) = -\mathcal{A}_q(a), \qquad \epsilon_{\mathcal{D}}(a) = a^{-3}\mathcal{A}_{\mathrm{w}}(a) + \mathcal{A}_q(a). \tag{16}$$

If the residual is separately conserved, then

$$\frac{\mathrm{d}\mathcal{A}_q}{\mathrm{d}\ln a} + a^{-3}\frac{\mathrm{d}\mathcal{A}_{\mathrm{w}}}{\mathrm{d}\ln a} = 0. \tag{17}$$

Thus the general separately conserved homogeneous residual is a one-function flow in $(\mathcal{A}_q, \mathcal{A}_{\mathrm{w}})$ space. The minimal dust-plus-vacuum branch is the constant nonnegative submanifold.

Theorem 3 (Minimal homogeneous residual closure)**.** *Let $I \subset \mathbb{R}_+$ be a connected scale-factor interval, and let $(\epsilon_{\mathcal{D}}, p_{\mathcal{D}}) \in C^1(I)$ be the homogeneous locally-FLRW Einstein residual after a fixed visible-sector subtraction. Then the residual is representable at the homogeneous stress-tensor level as*

$$T^{\mathcal{D}}_{\mu\nu} = \epsilon_{\mathrm{w},0} a^{-3}\frac{U_\mu U_\nu}{c^2} - \epsilon_q g_{\mu\nu}, \qquad \epsilon_{\mathrm{w},0}, \epsilon_q \geq 0, \tag{18}$$

if and only if $\mathcal{A}_q$ and $\mathcal{A}_{\mathrm{w}}$ are constant and nonnegative on I. In that case

$$\begin{aligned} \epsilon_q &= \mathcal{A}_q, \qquad \epsilon_{\mathrm{w},0} = \mathcal{A}_{\mathrm{w}}, \\ \epsilon_{\mathcal{D}}(a) &= \epsilon_{\mathrm{w},0} a^{-3} + \epsilon_q, \qquad p_{\mathcal{D}}(a) = -\epsilon_q. \end{aligned} \tag{19}$$

Proof. If Eq. (18) holds, then $-p_{\mathcal{D}} = \epsilon_q$ and $a^3(\epsilon_{\mathcal{D}} + p_{\mathcal{D}}) = \epsilon_{\mathrm{w},0}$. Conversely, if $\mathcal{A}_q$ and $\mathcal{A}_{\mathrm{w}}$ are constant, then Eq. (16) gives $p_{\mathcal{D}} = -\mathcal{A}_q$ and $\epsilon_{\mathcal{D}} = \mathcal{A}_{\mathrm{w}} a^{-3} + \mathcal{A}_q$, which is precisely Eq. (18). Nonnegativity is exactly the condition $\epsilon_q, \epsilon_{\mathrm{w},0} \geq 0$. □

The "if and only if" is an iff statement for the homogeneous stress tensor. A microscopic branch realizes the constants only if its charge, mass scale, flux normalization, branch labels, and quotient volume admit those amplitudes. In particular, a discrete flux lattice can make some otherwise admissible constant values of ϵ_q unavailable unless the flux unit or branch spectrum is correspondingly specified.

The invariant null tests are

$$\mathcal{R}_q := \frac{\mathrm{d}\mathcal{A}_q}{\mathrm{d}\ln a} = 0, \qquad \mathcal{R}_{\mathrm{w}} := \frac{\mathrm{d}\mathcal{A}_{\mathrm{w}}}{\mathrm{d}\ln a} = 0, \qquad \mathcal{A}_q, \mathcal{A}_{\mathrm{w}} \geq 0. \tag{20}$$

A failure of these tests falsifies the minimal homogeneous branch under the adopted visible subtraction; it does not falsify compact topology. Since the closed minimal

branch is separately conserved, Eq. (6) also implies separate conservation of the declared visible sector in the same convention. Failure of visible-sector conservation is therefore already a failure of the adopted minimal-source decomposition, not an independent topology effect.

Corollary 4 (Local ΛCDM degeneracy)**.** *On any local FLRW chart, the minimal branch is exactly the stress tensor of a pressureless component plus a cosmological-constant component with the same amplitudes and curvature. No homogeneous observable depending only on $H(a)$, distances, and the same curvature can distinguish the minimal branch from CDM plus Λ. A CDM plus Λ model placed on the same compact quotient, with the same projected perturbation covariance, the same initial data, and the same amplitudes, remains degenerate with the minimal branch at this level.*

V. CURRENT AND FOUR-FORM REPRESENTATION

The minimal source representation consists of a pressureless conserved-current branch and a fixed four-form branch. The current branch may carry a protected degree label, but topology alone does not give a dust stress tensor.

The degree label is standard in algebraic topology [4], while the pressureless current representative is a Brown-fluid effective description [5]. Let Σ be compact and oriented, let $\phi : \Sigma \to S^3$, and normalize the target volume form by $\int_{S^3} \widehat{\Omega} = 1$. The degree is

$$Q = \int_\Sigma \phi^* \widehat{\Omega} \in \mathbb{Z}. \tag{21}$$

The associated topological current is identically conserved, but

$$Q \in \mathbb{Z} \not\Rightarrow N^\mu = n(a) u^\mu \not\Rightarrow p_{\rm w} = 0. \tag{22}$$

The dust result requires a homogeneous coarse-grained Brown-current representative and a linear energy density

$$\epsilon_{\rm w}(n) = \mu_{\rm w} c^2 n. \tag{23}$$

More generally, a barotropic current effective theory with $\epsilon_J = F(n)$ has

$$p_J = nF'(n) - F(n), \qquad \frac{v_{s,J}^2}{c^2} := \frac{\mathrm{d}p_J}{\mathrm{d}\epsilon_J} = \frac{nF''(n)}{F'(n)}. \tag{24}$$

The minimal pressureless branch is the special case $F(n) = \mu_{\rm w} c^2 n$, for which $p_J = 0$ and $v_{s,J} = 0$.

For a homogeneous coarse-grained current branch on $\Sigma_{A_0} = S^3_{A_0}/\Gamma$,

$$\mathrm{Vol}(\Sigma_{A_0}) = \frac{2\pi^2 A_0^3}{|\Gamma|}, \qquad n(a) = \frac{|Q|}{a^3 \,\mathrm{Vol}(\Sigma_{A_0})}, \tag{25}$$

and therefore

$$\Omega_{\rm w} = \frac{\mu_{\rm w} c^2 |Q|}{\mathrm{Vol}(\Sigma_{A_0}) \epsilon_{\mathrm{crit},0}} = \frac{4G\mu_{\rm w}|Q|\,|\Gamma|}{3\pi H_0^2 A_0^3}. \tag{26}$$

This is an exact bookkeeping identity once $(Q, \mu_{\rm w}, |\Gamma|, A_0)$ are supplied. It is not an amplitude prediction.

Four-form vacuum sectors of this kind are standard in gravitational theory [6–9]. For a four-form $F_4 = \mathrm{d}C_3$, take

$$S_F = -\frac{Z_F}{2 \cdot 4!} \int \sqrt{-g}\, F_{\mu\nu\rho\sigma} F^{\mu\nu\rho\sigma}\, \mathrm{d}^4x, \qquad Z_F > 0. \tag{27}$$

The field equation implies that the four-form is proportional to the metric volume form on each connected branch,

$$F_{\mu\nu\rho\sigma} = f \varepsilon_{\mu\nu\rho\sigma}, \qquad f = \text{constant}, \tag{28}$$

up to the orientation convention for the Hodge dual. Only f^2 enters the stress tensor. Define

$$\widehat{q} = \sqrt{Z_F} f, \qquad \epsilon_q = \frac{1}{2}\widehat{q}^2. \tag{29}$$

The stress tensor obtained from Eq. (27) is

$$T^F_{\mu\nu} = \frac{Z_F}{3!} F_{\mu\alpha\beta\gamma} F_\nu{}^{\alpha\beta\gamma} - \frac{Z_F}{2 \cdot 4!} g_{\mu\nu} F_{\alpha\beta\gamma\delta} F^{\alpha\beta\gamma\delta}. \tag{30}$$

With the $(-+++)$ signature,

$$\varepsilon_{\mu\alpha\beta\gamma} \varepsilon_\nu{}^{\alpha\beta\gamma} = -3! g_{\mu\nu}, \qquad F_{\alpha\beta\gamma\delta} F^{\alpha\beta\gamma\delta} = -4! f^2, \tag{31}$$

which gives

$$T^q_{\mu\nu} = -\epsilon_q g_{\mu\nu}, \qquad \Lambda_{\rm eff} = \frac{8\pi G}{c^4} \epsilon_q. \tag{32}$$

For a fixed superselection branch, the mixed-index perturbation vanishes,

$$\delta T^{q\,\mu}{}_\nu = 0, \tag{33}$$

while the covariant-index variation

$$\delta T^q_{\mu\nu} = -\epsilon_q \delta g_{\mu\nu} \tag{34}$$

is purely metric-induced.

Proposition 5 (Closure versus flux-lattice selection)**.** *The homogeneous closure criterion determines only the stress-energy form*

$$T^q_{\mu\nu} = -\epsilon_q g_{\mu\nu}, \qquad \epsilon_q = \textit{constant} \geq 0. \tag{35}$$

It does not by itself determine whether the constant belongs to a discrete flux lattice. If a compact zero-mode normalization is supplied so that

$$\widehat{q} = N_F e_F, \qquad N_F \in \mathbb{Z}, \tag{36}$$

then the admissible vacuum amplitudes are restricted to

$$\epsilon_q = \frac{1}{2} N_F^2 e_F^2. \tag{37}$$

Thus the FLRW closure theorem is a continuous stress-energy classification, whereas flux quantization is an additional amplitude-selection constraint. Arbitrary residual matching requires an unfixed normalization e_F, a dense landscape of allowed branches, or a phenomenological effective constant.

VI. COMPACT TOPOLOGY AND QUOTIENT PROJECTION

The spatial topology is a global input, not a local equation of state. Throughout the topological-current and compact three-form slots we restrict to compact orientable spherical space forms, equivalently to finite freely acting orientation-preserving isometry groups Γ on S^3. Without orientability the degree and top-cohomology statements must be replaced by their twisted-coefficient versions; such branches are not considered here. The classification of complete constant-curvature space forms gives [2, 3]

$$\Sigma = \widetilde{\Sigma}/\Gamma, \qquad \widetilde{\Sigma} \in \{S^3, \mathbb{R}^3, H^3\}. \tag{38}$$

In the compact positive-curvature case,

$$\Sigma = S^3_A/\Gamma, \qquad \mathrm{Vol}(\Sigma) = \frac{2\pi^2 A^3}{|\Gamma|}. \tag{39}$$

If simple connectivity is assumed, $\Gamma = \mathbf{1}$ and $\Sigma \simeq S^3_A$. Without simple connectivity, the invariant conclusion is only that the universal cover is S^3_A.

On the simply connected cover,

$$-\Delta_{S^3_A} Y_{n\ell m} = \frac{n(n+2)}{A^2} Y_{n\ell m}, \qquad \mathrm{mult}(n) = (n+1)^2. \tag{40}$$

For a quotient, fields are Γ-invariant cover fields. The projector is

$$\boxed{P_\Gamma = \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \gamma^*.} \tag{41}$$

The physical scalar eigenspace is $\mathcal{H}_n^\Gamma = P_\Gamma \mathcal{H}_n$, with multiplicity

$$m_\Gamma(n) = \mathrm{Tr}_{\mathcal{H}_n} P_\Gamma. \tag{42}$$

The same logic applies to physical vector and tensor harmonic sectors after closed-FLRW gauge and constraint modes are removed.

Compact topology can therefore enter through

$$\mathrm{Vol}(\Sigma), \qquad \mathrm{Spec}_\Gamma(\Delta), \qquad m_\Gamma(n), \qquad \mathrm{Cov}_\Gamma, \tag{43}$$

not through a new local homogeneous stress tensor. Observer-position dependence appears for non-globally-homogeneous quotients; for globally homogeneous quotients the covariance is still quotient-projected but is not position dependent in the same way.

Criterion 6 (Topology-evidence boundary)**.** *A compact spherical model may claim compact-background consistency after specifying S^3/Γ and the quotient projection. It may not claim topology evidence or benchmark superiority until the projector has been propagated into transfer functions and observable covariance,*

$$P_\Gamma \longrightarrow m_\Gamma(n),\ \Xi_n^\Gamma,\ \Delta_{\ell n}^{X,\Gamma},\ C_\ell^{XY,\Gamma},\ \mathrm{Cov}_\Gamma,\ \mathcal{L}_\Gamma. \tag{44}$$

VII. AMPLITUDE-SELECTION BOUNDARY

Compact topology supplies labels, volume, and spectral restrictions. It does not supply numerical cosmological abundances. In particular,

$$(\Gamma, [\Sigma, S^3], H^3(\Sigma, \mathbb{Z}), Q, N_F) \not\Rightarrow (\Omega_{\mathrm{w}}, \Omega_q, \Lambda_{\mathrm{eff}}). \tag{45}$$

Even after the geometric radius A_0 is supplied, the amplitudes require μ_{w}, e_F, branch labels, compact-gauge normalization, and a selection rule or phenomenological prior. This is an amplitude-selection boundary, not a failure of the closure theorem.

A useful diagnostic for selector strength is the curvature-linear ansatz $\Omega_{\mathrm{w}} = C|\Omega_K|$. If observations impose $|\Omega_K| < \kappa$ and the target is $\Omega_\star$, then

$$C > \frac{\Omega_\star}{\kappa}. \tag{46}$$

For $\Omega_\star \simeq 0.26$ and $\kappa = 0.005$, one needs $C > 52$. A toy holographic winding saturation condition,

$$\mu_{\mathrm{w}} c^2 |Q| = \frac{c^4 A_0}{2G}, \tag{47}$$

gives only

$$\Omega_{\mathrm{w}}^{\mathrm{HWS}} = \frac{2}{3\pi} |\Omega_K| \tag{48}$$

for the simply connected case. For a quotient, imposing the same total-energy saturation on the quotient volume gives

$$\Omega_{\mathrm{w}}^{\mathrm{HWS}}(\Gamma) = \frac{2|\Gamma|}{3\pi} |\Omega_K|. \tag{49}$$

This is not a derivation of Ω_c. If $|\Gamma| \sim 10^2$ is used to obtain a large effective coefficient, the proposal becomes a large-quotient amplitude ansatz that must pass matched-circle, observer-position, covariance, and spectral-multiplicity tests.

VIII. PERTURBATIVE AND OBSERVATIONAL NULL TESTS

We use standard gauge-invariant cosmological perturbation conventions [10–13]. At linear order, the residual is defined before gauge choice by

$$\delta T^{\mathcal{D}}_{\mu\nu} = \frac{c^4}{8\pi G} \delta G_{\mu\nu} - \delta T^{\mathrm{vis}}_{\mu\nu}. \tag{50}$$

The fixed four-form branch contributes no independent scalar perturbation on a fixed branch. After subtracting this fixed vacuum component, the minimal current branch must obey the dust null conditions on every physical quotient-projected scalar mode:

$$\boxed{\delta p_{\mathrm{w,rf}} = 0, \qquad \Pi_{\mathrm{w}}^{\mathrm{GI}} = 0, \qquad \frac{v_{s,\mathrm{w}}^2}{c^2} = 0.} \tag{51}$$

A statistically significant intrinsic pressure perturbation, rest-frame sound speed, anisotropic stress, entropy source, or independent non-dust stress source not assigned to the visible sector falsifies the minimal perturbative branch.

Current background data sharpen the closure tests without changing their logical status. Planck 2018 combined with BAO gives a curvature constraint consistent with flatness, conventionally quoted as $\Omega_K = 0.0007 \pm 0.0019$ for TT,TE,EE+lowE+lensing+BAO [16]. Recent DESI DR2 BAO results are well described by flat ΛCDM, while BAO+CMB prefers a $w_0 w_a$ model over ΛCDM at 3.1σ, and BAO+CMB plus supernova samples gives a sample-dependent preference in the range 2.8–4.2σ [18]. If a reconstructed residual pressure evolution remains significant after systematics, dataset tension, and visible-sector convention dependence are accounted for, then the fixed-four-form minimal branch fails the $\mathcal{R}_q = 0$ null test. That would falsify the fixed-vacuum source law, not compact spherical topology.

Likewise, matched-circle and compact-covariance limits constrain detectable domains and topology families, not every compact spherical quotient. Lens-space constraints are therefore constraints on specified S^3/Γ implementations and their observer-position/covariance pipelines, where applicable [14, 15, 17].

IX. CONCLUSION

We have proved a closure theorem for homogeneous Einstein residuals on compact spherical FLRW backgrounds. After the metric, Einstein-side convention, and visible-sector subtraction are fixed, the residual belongs to the minimal pressureless-current plus fixed-four-form branch exactly when

$$-p_{\mathcal{D}} = \text{constant} \geq 0, \qquad a^3(\epsilon_{\mathcal{D}} + p_{\mathcal{D}}) = \text{constant} \geq 0. \tag{52}$$

Equivalently,

$$\epsilon_{\mathcal{D}}(a) = \epsilon_{\mathrm{w},0} a^{-3} + \epsilon_q, \qquad p_{\mathcal{D}}(a) = -\epsilon_q. \tag{53}$$

This proves local background equivalence to cold dark matter plus Λ with the same curvature and amplitudes.

Compact topology can still enter through finite volume, quotient-projected spectra, compact covariance, and observer-position effects for non-globally-homogeneous quotients. It does not, however, determine Ω_c, Ω_Λ, or Λ_{eff}. The Brown-current dust behavior requires a coarse-grained pressureless current EFT; it is not a consequence of a winding integer alone. Likewise, a fixed four-form supplies a vacuum-like stress tensor, but its flux spacing and branch selection require additional data.

The value of the theorem is not that it produces a new homogeneous expansion history, but that it gives a convention-explicit null test for when a reconstructed Einstein residual may consistently be interpreted as a pressureless protected current plus fixed four-form vacuum, while proving that compact topology alone cannot select the observed amplitudes. To become a topology-evidence claim it must be supplemented by quotient transfer functions, compact covariance kernels, likelihoods, and blind-injection tests.

[1] R. M. Wald, *General Relativity* (University of Chicago Press, Chicago, 1984).
[2] S. W. Hawking and G. F. R. Ellis, *The Large Scale Structure of Space-Time* (Cambridge University Press, Cambridge, 1973).
[3] J. A. Wolf, *Spaces of Constant Curvature*, 6th ed. (AMS Chelsea, Providence, 2011).
[4] A. Hatcher, *Algebraic Topology* (Cambridge University Press, Cambridge, 2002).
[5] J. D. Brown, Action functionals for relativistic perfect fluids, Class. Quantum Grav. **10**, 1579–1606 (1993).
[6] A. Aurilia, H. Nicolai, and P. K. Townsend, Hidden constants: the theta parameter of QCD and the cosmological constant of $N = 8$ supergravity, Nucl. Phys. B **176**, 509 (1980).
[7] M. Henneaux and C. Teitelboim, The cosmological constant as a canonical variable, Phys. Lett. B **143**, 415 (1984).
[8] J. D. Brown and C. Teitelboim, Neutralization of the cosmological constant by membrane creation, Nucl. Phys. B **297**, 787 (1988).
[9] R. Bousso and J. Polchinski, Quantization of four-form fluxes and dynamical neutralization of the cosmological constant, JHEP **06**, 006 (2000).
[10] J. M. Bardeen, Gauge-invariant cosmological perturbations, Phys. Rev. D **22**, 1882 (1980).
[11] H. Kodama and M. Sasaki, Cosmological perturbation theory, Prog. Theor. Phys. Suppl. **78**, 1 (1984).
[12] V. Mukhanov, H. Feldman, and R. Brandenberger, Theory of cosmological perturbations, Phys. Rep. **215**, 203 (1992).

[13] C.-P. Ma and E. Bertschinger, Cosmological perturbation theory in the synchronous and conformal Newtonian gauges, Astrophys. J. **455**, 7 (1995).
[14] N. J. Cornish, D. N. Spergel, G. D. Starkman, and E. Komatsu, Constraining the topology of the Universe, Phys. Rev. Lett. **92**, 201302 (2004).
[15] P. A. R. Ade *et al.* (Planck Collaboration), Planck 2015 results. XVIII. Background geometry and topology, Astron. Astrophys. **594**, A18 (2016), arXiv:1502.01593.
[16] N. Aghanim *et al.* (Planck Collaboration), Planck 2018 results. VI. Cosmological parameters, Astron. Astrophys. **641**, A6 (2020), arXiv:1807.06209.
[17] S. Saha *et al.* (COMPACT Collaboration), Cosmic topology. Part Ic. Limits on lens spaces from circle searches, JCAP **01**, 004 (2025), arXiv:2409.02226.
[18] M. Abdul Karim *et al.* (DESI Collaboration), DESI DR2 results II: Measurements of baryon acoustic oscillations and cosmological constraints, Phys. Rev. D **112**, 083515 (2025), arXiv:2503.14738.